VSL-R1: What Makes Extra Preparation Time Useful?

Conditional transfer-clock and receiving audit

Method-first task-definition review

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Status and purpose. These are analytical audit deductions, with no simulation or new primary traffic law adopted. S13 remains the accepted conditional reference. The audit stops at gate VSL-R1-G1 because a fixed physical-time transfer rate and a flow/distance-coupled transfer are different models under common-speed control.

1. The marginal benefit the inherited task needs

Keep S13's fixed affected mass $M > 0$ in veh-eq, zone length $L > 0$ in metres, common realized speed $v > 0$ in m/s, receiving-work factor $\alpha > 1$ and work rate $\mu > 0$ in normalized veh-eq/s. The ideal traversal time is $\tau = L/v$. Let $E(v)$ be the mass remaining in the ending lane at the choke. Conditional on the *same proportional batch-service law*, the mean complete time is

$$J(v) = \frac{L}{v} + \frac{M}{2\mu} + aE(v), \qquad a = \frac{\alpha - 1}{2\mu}. \tag{1}$$

J is seconds; a has units s/veh-eq. Total time is $F = MJ$, in veh-eq s. Consequently, in a smooth branch,

$$J'(v) = -\frac{L}{v^2} + aE'(v). \tag{2}$$

A small feasible slowdown $\delta v < 0$ improves J to first order only if

$$J'(v) > 0 \quad \Longleftrightarrow \quad aE'(v) > \frac{L}{v^2}. \tag{3}$$

The first term is the added travel time everyone pays. The second is the reduction in mean service burden under this particular allocation. Finite receiving and positive lateral transfers alone do not establish the inequality. At the fastest bound the slowdown is feasible; an increase beyond the bound is not. Equality gives no first-order conclusion.

This expression must not be reused unchanged if preparation interaction delay, arrival chronology, receiving or allocation is altered. Those changes require their own complete departure-time accounting. The fixed coefficient a is not a source-independent service identity.

2. Holding a time rate fixed is a scientific assumption

For a diagnostic, suppose a nonbinding, prescribed-motion preparation uses a fractional completion rate $\lambda(v) > 0$ in s^{-1}, constant along one traversal at a fixed action. With initial ending mass $E_0 = pM$,

$$E(v) = E_0 \exp\left[-\lambda(v)\frac{L}{v}\right]. \tag{4}$$

This diagnostic family is **not adopted as a new traffic model**. It exposes what a source-to-reduction argument would have to justify. Direct differentiation gives

$$E'(v) = \frac{LE(v)}{v^2}\big[\lambda(v) - v\lambda'(v)\big], \tag{5}$$

$$J'(v) = \frac{L}{v^2}\left[-1 + aE(v)\big(\lambda(v) - v\lambda'(v)\big)\right]. \tag{6}$$

The speed dependence of the rate matters as much as its positive value.

Inherited S13 clock. Hold the physical-time rate fixed: $\lambda(v) = \lambda_0 > 0$. Then

$$E(v) = E_0 e^{-\lambda_0 L/v}, \qquad J'(v) = \frac{L}{v^2}\big[-1 + a\lambda_0 E(v)\big]. \tag{7}$$

This restates the accepted sensitivity and its conditional improving regime.

Speed-scaled countercase. Hold a per-distance completion coefficient $\beta > 0$ in m^{-1} fixed, so $\lambda(v) = \beta v$. Then

$$E(v) = E_0 e^{-\beta L}, \qquad E'(v) = 0, \qquad J'(v) = -L/v^2 < 0. \tag{8}$$

Longer residence is exactly offset by a smaller per-time rate. Under the inherited service law, fastest speed is best for every preparation even though conservative merging and extra receiving work remain present.

Source connection and limit. FB0030's equation 14 makes desired lateral flow a fraction of sending demand; its free-flow sending branch (equation 13) contains speed times density. That motivates checking rate scaling. Equation (8) is an original conditional diagnostic, not a continuum derivation or VSL prediction of the full native two-step model. Density, incentives, receiving and cooperation may break this scaling. Carey's location-based desire likewise is not a prescribed realized-completion schedule. Neither source identifies $\lambda'(v) = 0$ for this task.

3. Conditional receiving reduction behind batch co-clearance

The audited Carey R2 allocation uses gap-weighted receiving demands. Here we project that interpretation onto two simultaneously available queues, rather than claiming to reproduce the source's spatial update system.

At the start of service let $e(0) = E$, $c(0) = M - E$. For remaining masses c, e, assume constant work rate μ, unit work for continuing mass and α work for ending mass. All demand is exposed, with no other binding constraint. Proportional allocation to work demands c and αe gives

$$\dot{c} = -\frac{\mu c}{c + \alpha e}, \qquad \dot{e} = -\frac{\mu e}{c + \alpha e}, \qquad c + e > 0. \tag{9}$$

For positive classes their ratio is invariant; zero classes remain zero. Writing $B = c + e$, $f = E/M$ and $\overline{w} = 1 + (\alpha - 1)f$ yields

$$c + \alpha e = \overline{w} B, \qquad \dot{B} = -\mu/\overline{w}, \qquad T = \frac{\overline{w} M}{\mu} = \frac{M + (\alpha - 1)E}{\mu}. \tag{10}$$

Thus, for $0 \le u \le T$,

$$B(u) = M(1 - u/T), \qquad \int_0^T B(u)\,\mathrm{d}u = \frac{MT}{2}. \tag{11}$$

Service stops at $B = 0$; the quotient is not evaluated there. Work is consumed at rate μ, and discharged traffic is the removed mass, not the work. Including the common preparation traversal gives $F = M\tau + MT/2$, as in S13.

This is a coherent conditional resource model. It requires a simultaneous batch, constant accessible receiving, the stated allocation and no independent donor/sending limitation. A spatial CTM generally does not present its entire population as those two queue demands. Sharing a target cell is also different from assuming both continuing lanes offer one accessible resource.

Preparation receiving. Nonbinding preparation is a separate regime assumption. A lane change can consume receiving space without adding travel delay when that resource has slack, but that slack must be established over the action range. Slower motion can change density and slack. The choke projection does not provide such a regime bound or a physical-time completion law.

Information and gate. E_0 alone still reproduces the inherited grouped model under its fixed law. No future reference $E(v)$ is supplied. Native receiving-limited extensions may need adjacent-lane state; none is selected here. VSL-R1-G1 concerns retaining an explicit conditional benchmark versus amending its primary preparation/receiving contract. Independent useful traffic gradients, calibration and implementation checks remain unestablished.